# Lecture 3 - Limits II

Garrett Credi

09/14/2026

# Announcements

Homework Policy:

# Announcements

Homework Policy:

- For homework assigned on Tueday, the due date will be Thursday @ 6PM

# Announcements

Homework Policy:

- For homework assigned on Tueday, the due date will be Thursday @ 6PM
- For homework assigned on Thursday, the due date will be Saturday @ 6PM

# Announcements

Homework Policy:

- For homework assigned on Tueday, the due date will be Thursday @ 6PM
- For homework assigned on Thursday, the due date will be Saturday @ 6PM

Office Hours:

# Announcements

Homework Policy:

- For homework assigned on Tueday, the due date will be Thursday @ 6PM
- For homework assigned on Thursday, the due date will be Saturday @ 6PM

Office Hours:

- Wednesday 12:15 - 1:15 PM
- Thursday 4:30 - 5:30 PM

# Announcements

Homework Policy:

- For homework assigned on Tueday, the due date will be Thursday @ 6PM
- For homework assigned on Thursday, the due date will be Saturday @ 6PM

Office Hours:

- Wednesday 12:15 - 1:15 PM
- Thursday 4:30 - 5:30 PM

Both are in Vincent Hall 105.

# Last Time:

Last lecture we:

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function:

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:
    - points where the graph of a function “broke” were points where the limit did not exist

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:
   - points where the graph of a function "broke" were points where the limit did not exist
   - points where the graph of a function had a "hole" had the limit exist but it ***was not equal to the function's value***

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:
   - points where the graph of a function "broke" were points where the limit did not exist
   - points where the graph of a function had a "hole" had the limit exist but it ***was not equal to the function's value***
   - points where the graph was just an unbroken curve had the limit exist and *equal* to the value at that point

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:
   - points where the graph of a function "broke" were points where the limit did not exist
   - points where the graph of a function had a "hole" had the limit exist but it ***was not equal to the function's value***
   - points where the graph was just an unbroken curve had the limit exist and *equal* to the value at that point
4. *Approximated* limits using a table of values

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:
   - points where the graph of a function "broke" were points where the limit did not exist
   - points where the graph of a function had a "hole" had the limit exist but it ***was not equal to the function's value***
   - points where the graph was just an unbroken curve had the limit exist and *equal* to the value at that point
4. *Approximated* limits using a table of values
5. Computed limits by algebraic manipulation

# Today's Goals:

Today we will develop our ability to algebraically compute limits by:

# Today's Goals:

Today we will develop our ability to algebraically compute limits by:

1. Studying the "simplest" functions to take limits of, *continuous functions*
2. Discovering *limit properties* that help us reduce complicated limits to easier limits

# Today's Goals:

Today we will develop our ability to algebraically compute limits by:

1. Studying the "simplest" functions to take limits of, *continuous functions*
2. Discovering *limit properties* that help us reduce complicated limits to easier limits
3. Combining the above to get the process of algebraically computing limits

And we will also study cases where the limit is infinite.

# Continuity

# Continuous Functions I

To start off: let's consider the function:

$$f(x) = \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} \tag{1}$$

# Continuous Functions I

To start off: let's consider the function:

$$f(x) = \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} \tag{1}$$

1. What is the domain of $f$?

# Continuous Functions I

To start off: let's consider the function:

$$f(x) = \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} \tag{1}$$

1. What is the domain of $f$?
2. Using the graph of $f(x)$, what are:
   (1) $\lim_{x \to 3} f(x)$?
   (2) $\lim_{x \to 5} f(x)$?
   (3) $\lim_{x \to 1} f(x)$?

# Continuous Functions I

To start off: let's consider the function:

$$f(x) = \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} \tag{1}$$

1. What is the domain of $f$?
2. Using the graph of $f(x)$, what are:
   (1) $\lim_{x \to 3} f(x)$?
   (2) $\lim_{x \to 5} f(x)$?
   (3) $\lim_{x \to 1} f(x)$?

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

Continuity at a point

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

## Continuity at a point

Given a function $f(x)$ and a real number $a$ ,

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

**Continuity at a point**

Given a function $f(x)$ and a real number $a$ , we say that $f(x)$ *is continuous at* $a$ if:

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

## Continuity at a point

Given a function $f(x)$ and a real number $a$ , we say that $f(x)$ *is continuous at* $a$ if:

1. The limit $\lim_{x \to a} f(x)$ exists

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

## Continuity at a point

Given a function $f(x)$ and a real number $a$ , we say that $f(x)$ *is continuous at* $a$ if:

1. The limit $\lim_{x \to a} f(x)$ exists
2. $a$ is in the domain of $f$

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

**Continuity at a point**

Given a function $f(x)$ and a real number $a$ , we say that $f(x)$ *is continuous at* $a$ if:

1. The limit $\lim_{x \to a} f(x)$ exists
2. $a$ is in the domain of $f$
3. $\lim_{x \to a} f(x) = f(a)$

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

Continuous on $[a, b]$

Continuous Function

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if,

**Continuous Function**

A function $f(x)$ is said to be continuous if,

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if, for every real number $a \leq x_0 \leq b$

**Continuous Function**

A function $f(x)$ is said to be continuous if,

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if, for every real number $a \leq x_0 \leq b$ we have that $f(x)$ is continuous at $x_0$.

**Continuous Function**

A function $f(x)$ is said to be continuous if,

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if, for every real number $a \leq x_0 \leq b$ we have that $f(x)$ is continuous at $x_0$.

**Continuous Function**

A function $f(x)$ is said to be continuous if,

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if, for every real number $a \leq x_0 \leq b$ we have that $f(x)$ is continuous at $x_0$.

**Continuous Function**

A function $f(x)$ is said to be continuous if, for every real number $x_0$ in the domain of $f$,

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if, for every real number $a \leq x_0 \leq b$ we have that $f(x)$ is continuous at $x_0$.

**Continuous Function**

A function $f(x)$ is said to be continuous if, for every real number $x_0$ in the domain of $f$, we have that $f(x)$ is continuous at $x_0$.

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits

Important Continuous Functions

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches

Important Continuous Functions

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.

Important Continuous Functions

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

Important Continuous Functions

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

1. The constant function $f_c(x) = c$ for some constant number $c$.

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

1. The constant function $f_c(x) = c$ for some constant number $c$.
2. The identity function $f_{id}(x) = x$

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

1. The constant function $f_c(x) = c$ for some constant number $c$.
2. The identity function $f_{id}(x) = x$
3. The exponential and logarithm functions $\exp(x) = e^x, \ln(x)$

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

1. The constant function $f_c(x) = c$ for some constant number $c$.
2. The identity function $f_{id}(x) = x$
3. The exponential and logarithm functions $\exp(x) = e^x, \ln(x)$
4. All trigonometric functions, $\sin(x), \cos(x), \tan(x)$...

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

1. The constant function $f_c(x) = c$ for some constant number $c$.
2. The identity function $f_{id}(x) = x$
3. The exponential and logarithm functions $\exp(x) = e^x, \ln(x)$
4. All trigonometric functions, $\sin(x), \cos(x), \tan(x)$...

# Continuous Functions V

What about other familiar functions like:

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,
2. Polynomials $c_0x^n + c_1x^{n-1} + \ldots + c_nx + c_{n+1}$,

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,
2. Polynomials $c_0 x^n + c_1 x^{n-1} + \ldots + c_n x + c_{n+1}$,
3. Rational functions $\dfrac{p(x)}{q(x)}$

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,
2. Polynomials $c_0x^n + c_1x^{n-1} + \ldots + c_nx + c_{n+1}$,
3. Rational functions $\dfrac{p(x)}{q(x)}$
4. Weird functions like $e^{\sin(x)} + \cos(x^n)\sqrt{x}$?

We can determine the points at which these functions are continuous

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,
2. Polynomials $c_0x^n + c_1x^{n-1} + \ldots + c_nx + c_{n+1}$,
3. Rational functions $\dfrac{p(x)}{q(x)}$
4. Weird functions like $e^{\sin(x)} + \cos(x^n)\sqrt{x}$?

We can determine the points at which these functions are continuous (and thus the points at which it is easy to calculate limits)

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,
2. Polynomials $c_0x^n + c_1x^{n-1} + \ldots + c_nx + c_{n+1}$,
3. Rational functions $\dfrac{p(x)}{q(x)}$
4. Weird functions like $e^{\sin(x)} + \cos(x^n)\sqrt{x}$?

We can determine the points at which these functions are continuous (and thus the points at which it is easy to calculate limits) by discovering *limit properties*!

# Limit Properties

# Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and

# Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

# Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$

# Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$
2. Property 2 says that as long as $x \approx 0$ then $g(x) \approx 0$

# Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$
2. Property 2 says that as long as $x \approx 0$ then $g(x) \approx 0$

So if $x \approx 0$ then $h(x) = f(x) + g(x) \approx 1 + 0 = 1$!

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x) + g(x))$ exists

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x) + g(x))$ exists and is equal to $b + c$.

# Limit Properties: Multiplicativity

**Multiplicativity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

Similarly:

**Limits of Multiplicative Inverses**

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists

Similarly:

## Limits of Multiplicative Inverses

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$

Similarly:

## Limits of Multiplicative Inverses

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists

Similarly:

## Limits of Multiplicative Inverses

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Similarly:

## Limits of Multiplicative Inverses

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists

Similarly:

## Limits of Multiplicative Inverses

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and $a$ is a real number so that

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and $a$ is a real number so that $\lim_{x \to a} f(x)$ exists,

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and $a$ is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number $b$,

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and $a$ is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number $b$, and that $b \neq 0$.

Then we have that $\lim_{x \to a} \dfrac{1}{f(x)}$ exists

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and $a$ is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number $b$, and that $b \neq 0$.

Then we have that $\lim_{x \to a} \dfrac{1}{f(x)}$ exists and is equal to $\dfrac{1}{b}$.

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $f(x)$ is *not* constantly equal to $b$ around $x = a$,

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $f(x)$ is *not* constantly equal to $b$ around $x = a$,
3. $\lim_{x \to b} g(x)$ exists

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $f(x)$ is *not* constantly equal to $b$ around $x = a$,
3. $\lim_{x \to b} g(x)$ exists and is equal to $c$

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $f(x)$ is *not* constantly equal to $b$ around $x = a$,
3. $\lim_{x \to b} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a} g(f(x)) = c$.

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

Combining Continuous Functions

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$.

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$. Then we have that:

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$
2. $f(x)g(x)$ is continuous at $x = a$

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$
2. $f(x)g(x)$ is continuous at $x = a$
3. If $g(a)$ and $\lim_{x \to a} g(x)$ are both nonzero then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$.

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$
2. $f(x)g(x)$ is continuous at $x = a$
3. If $g(a)$ and $\lim_{x \to a} g(x)$ are both nonzero then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$.
4. If $g(x)$ is instead continuous at $f(a)$ then $g(f(x))$ is continuous at $x = a$.

# Moral of the Story

The big upshot of this is the following:

**Strategy for Computing Limits**

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically

# Moral of the Story

The big upshot of this is the following:

**Strategy for Computing Limits**

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down

# Moral of the Story

The big upshot of this is the following:

**Strategy for Computing Limits**

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

# Moral of the Story

The big upshot of this is the following:

**Strategy for Computing Limits**

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

1. $f(x)$ is a continuous function

# Moral of the Story

The big upshot of this is the following:

**Strategy for Computing Limits**

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

1. $f(x)$ is a continuous function
2. and $a$ is in the domain of $f(x)$.

# Moral of the Story: Example

For instance:

$$\lim_{x \to 1} \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} =$$

# Moral of the Story: Example

For instance:

$$\lim_{x \to 1} \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} = \lim_{x \to 1} \frac{x^3 - x^2 - x - 1}{(x - 1)(x + 2)}$$

# Moral of the Story: Example

For instance:

$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2}=\lim_{x\to 1}\frac{x^3-x^2-x-1}{(x-1)(x+2)} \tag{2}$$
$$=\lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)}$$

# Moral of the Story: Example

For instance:

$$\lim_{x \to 1} \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} = \lim_{x \to 1} \frac{x^3 - x^2 - x - 1}{(x-1)(x+2)} \tag{2}$$

$$= \lim_{x \to 1} \frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{3}$$

$$= \lim_{x \to 1} \frac{(x-1)}{(x-1)} \frac{x^2-1}{x+2}$$

# Moral of the Story: Example

For instance:

$$\lim_{x \to 1} \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} = \lim_{x \to 1} \frac{x^3 - x^2 - x - 1}{(x-1)(x+2)} \tag{2}$$

$$= \lim_{x \to 1} \frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{3}$$

$$= \lim_{x \to 1} \frac{(x-1)}{(x-1)} \frac{x^2-1}{x+2} \tag{4}$$

$$= \lim_{x \to 1} \frac{x^2-1}{x+2}$$

# Moral of the Story: Example

For instance:

$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2} = \lim_{x\to 1}\frac{x^3-x^2-x-1}{(x-1)(x+2)} \tag{2}$$

$$= \lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{3}$$

$$= \lim_{x\to 1}\frac{(x-1)}{(x-1)}\frac{x^2-1}{x+2} \tag{4}$$

$$= \lim_{x\to 1}\frac{x^2-1}{x+2} \tag{5}$$

$$= \frac{(1)^2-1}{1+2}$$

# Moral of the Story: Example

For instance:

$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2} = \lim_{x\to 1}\frac{x^3-x^2-x-1}{(x-1)(x+2)} \tag{2}$$

$$= \lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{3}$$

$$= \lim_{x\to 1}\frac{(x-1)}{(x-1)}\frac{x^2-1}{x+2} \tag{4}$$

$$= \lim_{x\to 1}\frac{x^2-1}{x+2} \tag{5}$$

$$= \frac{(1)^2-1}{1+2} = \frac{1-1}{3}$$

# Moral of the Story: Example

For instance:

$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2}=\lim_{x\to 1}\frac{x^3-x^2-x-1}{(x-1)(x+2)} \tag{2}$$

$$=\lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{3}$$

$$=\lim_{x\to 1}\frac{(x-1)}{(x-1)}\frac{x^2-1}{x+2} \tag{4}$$

$$=\lim_{x\to 1}\frac{x^2-1}{x+2} \tag{5}$$

$$=\frac{(1)^2-1}{1+2}=\frac{1-1}{3}=\frac{0}{3}=0 \tag{6}$$

# One Sided Limits

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

One-Sided Limits

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches *a from the left*")

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$)

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$)

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.
- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the right*")

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.
- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the right*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$)

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.
- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the right*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$)

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.
- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the right*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x > a$.

# Relationship between Limits and One-Sided Limits

Questions:

# Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?

# Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?
2. If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ what can we say about $\lim_{x \to a} f(x)$?

# Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?
2. If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ what can we say about $\lim_{x \to a} f(x)$?
3. If $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$, what can we say about $\lim_{x \to a} f(x)$?

# Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?
2. If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ what can we say about $\lim_{x \to a} f(x)$?
3. If $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$, what can we say about $\lim_{x \to a} f(x)$?
4. If $\lim_{x \to a^-} f(x) = f(a) = \lim_{x \to a^+} f(x)$, what previously discussed definition does the function $f(x)$ and the point $a$ satisfy?

# Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?
2. If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ what can we say about $\lim_{x \to a} f(x)$?
3. If $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$, what can we say about $\lim_{x \to a} f(x)$?
4. If $\lim_{x \to a^-} f(x) = f(a) = \lim_{x \to a^+} f(x)$, what previously discussed definition does the function $f(x)$ and the point $a$ satisfy?

# Worksheet Time!

Let’s get into groups and work on the worksheet!

# End of Class!

That’s all folks!